"Habit Tracker Dashboard"

1 . Product Vision

Build a small single-page dashboard that lets users create daily habits (e.g., **"Drink 2 L water"**, **"Study React 30 min"**) and tick them off for **today**. The UI shows:

1. **A form** for adding new habits.
2. A **card for each habit** containing:
   - Habit name
   - A ✓ **toggle** button for "done today"
   - A live **streak counter** (consecutive days completed)
3. A **summary banner** that displays how many habits are finished out of the total and a simple progress bar.

You're modelling the first slice of a full habit-tracking app, focusing purely on state and events with `useState`.

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2 . Learning Objectives

Concept	Practice Point
Multiple pieces of state (`array`, `boolean`)	Manage a list of habit objects and their "done" status
Updating complex state immutably	Adding a habit, toggling "done", updating streak
Controlled form components	Habit-creation input field
Derived state & conditional rendering	Progress banner, disabled states, dynamic streak number
Event handlers & data identity (`ids`)	Identifying which habit to toggle/remove

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3 . Core Requirements

ID	Requirement
R-1	**File structure** contains `App.jsx`, `HabitForm.jsx`, `HabitCard.jsx`, `SummaryBanner.jsx`, `index.jsx`.
R-2	`App` holds a **`habits` array** in state; each habit object must track `{ id, title, doneToday, streak }`.
R-3	`HabitForm` is a controlled component with an `<input>` and "Add" button. Submitting appends a new habit to `habits`, auto-setting `doneToday: false`, `streak: 0`, and a unique `id`.
R-4	`HabitCard` receives a habit object plus two callbacks: `onToggleDone(id)` and `onRemove(id)`. It renders the title, streak, a **Toggle** button, and a **Remove (🗑)** button.

ID	Requirement
R-5	Toggling "done" flips `doneToday` **and** increments or resets `streak` appropriately: • if marking as done → `streak + 1` • if un-marking → `streak - 1` (min 0).
R-6	`SummaryBanner` derives completed / total counts from `habits` and displays a horizontal progress bar that fills proportionally.
R-7	All buttons must have basic focus/hover styles; the toggle button visually differs between completed vs. not.
R-8	The app compiles without warnings in a standard Vite-React setup. No third-party state libraries.

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4 . Stretch Goals (optional)

1. **Midnight reset** — on first load each day, automatically set `doneToday` to `false` for every habit (hint: compare stored date vs. `new Date().toDateString()`).
2. **Local persistence** — save the `habits` array to `localStorage` on each change and hydrate on initial render.
3. **Filter chips** — buttons to view *All* / *Done* / *Pending* habits.
4. **Celebration** — show a confetti emoji 🎉 in the summary banner when all habits are completed.

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5 . Handoff Checklist

- ☐ Repo named `react-habit-tracker-YOURNAME`.
- ☐ `README.md` with setup instructions and one screenshot or short GIF.
- ☐ Each component file contains a **brief description comment** explaining its role.
- ☐ No ESLint or compile errors; runs with `npm run dev`.

> **Submission:** zip or push the repo before the next lecture. **Tip:** Use React DevTools to watch how the `habits` array evolves as you add and toggle items.

Build good habits—both in code **and** daily life! 🚀